

## Facial Recognition

An airport facial recognition is one of the biometric screenings that is widely used in the United States and other countries around the world. At the airport, the facial recognition is to compare the traveler that stands in front of the camera with photograph from their ID or passport. Biometric facial comparison technology is used when the traveler enters and exits the country, goes through TSA security check, and drops off their bags. (*Biometrics Environments: Airports*, 2018) The main purpose is to verify a traveler's identity more quickly and efficiently.

An airport facial recognition uses a computer vision and an AI to analyze the traveler's face. According to (*CBP, Lawa Expand Biometric Traveler Experience at LAX*, 2025), "CBP's biometric facial matching service will match the new photo of the traveler to a small gallery of high-quality images that the traveler has already provided to the government, such as passport and visa photos". The technology behind the airport facial recognition kiosk starts with a camera capture the traveler's face. The camera acts as the biometric sensor for traveler's identification. Then, the system detects the facial structure and analyzes it. The system detects and analyzes by using machine learning models. Finally, the system compares facial structures with provided images to determine if it is a match.

An airport facial recognition has many benefits to country security system. It also has many challenges and limitations. One of the benefits of airport facial recognition is that it is faster and more efficient way to verify traveler's identity in a shorter processing time. The challenges that the airport might face are that the system might miss match travelers with the image. The camera at the TSA kiosk might affect accuracy in the system because of the lighting and camera quality. Also, many people might be concerned about their privacy. It is one of the ethical challenges that they are facing. People might be concerned about how their biometric information is stored, collected and used without their consent.

I believe in the future airport facial recognition technology could be more accurate and used in every airport around the world. It could have major impact on travelers and people who work at the airports. In the future, when better camera could be more reasonable price to make a better kiosk camera, it could make airport security faster, easier and more accurate. If the system becomes more reliable, it could reduce the number of employees needed for TSA tasks. This could have a negative impact on airport workers because their jobs could be replaced by facial recognition system.

## Work Cites

*Biometrics environments: Airports*. (2018). U.S. Customs and Border Protection. <https://www.cbp.gov/travel/biometrics/environments/airports>

*CBP, Lawa expand biometric traveler experience at LAX*. (2025). U.S. Customs and Border Protection. [https://www.cbp.gov/newsroom/local-media-release/cbp-lawa-expand-biometric-traveler-experience-lax?utm\\_source=chatgpt.com](https://www.cbp.gov/newsroom/local-media-release/cbp-lawa-expand-biometric-traveler-experience-lax?utm_source=chatgpt.com)